Notes: 'n' represents the dimension of the corresponding problem. 'NI' represents the total number of iterations. 'NF' represents the number of the function evaluation. 'T' represents the running time of the algorithm. '$\|g^{\star}\|$' represents the gradient value for the solution. The values of 'NI', 'NF', 'T' and '$\|g^{\star}\|$' are set as 'NaN' if and only if the number of iterations is greater than 3000.

Table 1: Numerical results for Test 2

Problem/n	SHCG-V1 NI/NF/T/$\|g^{\star}\|$	SHCG-V3 NI/NF/T/$\|g^{\star}\|$	CG-Descent NI/NF/T/$\|g^{\star}\|$	FhDY NI/NF/T/$\|g^{\star}\|$	MP+ NI/NF/T/$\|g^{\star}\|$	SMABFGS NI/NF/T/$\|g^{\star}\|$
dixmaana/3000	19/83/0.088/1.96e-07	21/81/0.058/9.15e-07	24/77/0.052/6.47e-07	19/83/0.057/7.02e-08	25/80/0.057/2.43e-07	45/96/0.068/1.69e-07
dixmaana/30000	17/86/0.471/3.56e-07	20/86/0.469/9.75e-07	31/88/0.491/8.56e-07	19/86/0.474/3.86e-07	28/87/0.484/3.87e-07	82/135/0.781/8.66e-07
dixmaana/600000	22/97/10.919/9.49e-07	14/84/12.226/6.03e-07	24/92/21.969/2.24e-07	17/90/22.490/6.14e-07	22/86/19.390/9.81e-07	92/142/33.144/3.12e-07
dixmaanb/6000	21/88/0.230/2.93e-07	18/84/0.170/6.07e-07	27/83/0.177/5.87e-07	16/82/0.177/2.89e-07	22/78/0.214/6.67e-07	49/101/0.272/5.24e-07
dixmaanb/15000	26/93/0.642/5.66e-07	20/87/0.628/8.58e-08	27/79/0.517/9.37e-07	22/90/0.534/4.55e-09	23/73/0.512/3.05e-07	72/125/0.755/3.71e-07
dixmaanb/60000	19/89/2.038/9.11e-07	23/93/2.235/6.05e-07	27/90/2.021/3.75e-07	23/98/2.478/9.52e-07	25/80/1.997/9.71e-07	75/131/3.197/3.00e-08
dixmaanc/3000	21/80/0.125/8.82e-07	18/81/0.085/9.54e-07	26/82/0.085/3.01e-07	19/84/0.086/5.05e-07	19/75/0.078/5.81e-08	40/86/0.086/2.22e-07
dixmaanc/150000	23/97/5.542/6.40e-07	21/92/5.171/9.06e-07	21/85/5.323/4.33e-07	20/95/5.712/6.21e-07	26/91/5.347/9.73e-07	90/146/9.331/6.97e-07
dixmaand/300000	15/86/7.407/3.48e-07	19/91/4.938/6.66e-07	27/93/5.015/9.15e-07	17/80/4.301/8.83e-07	28/93/5.051/1.66e-07	74/128/6.972/5.07e-07
dixmaane/1500	297/413/0.166/7.81e-07	315/399/0.133/6.24e-07	565/661/0.253/8.38e-07	308/589/0.221/9.87e-07	687/761/0.275/9.05e-07	382/430/0.140/5.60e-07
dixmaane/3000	431/601/0.393/9.98e-07	478/583/0.372/8.63e-07	793/911/0.589/9.52e-07	433/802/0.511/9.47e-07	761/851/0.549/7.66e-07	665/726/0.477/4.64e-07
dixmaane/15000	820/1054/3.203/8.37e-07	825/1039/3.199/7.91e-07	1907/2192/6.663/8.86e-07	753/1464/4.460/9.06e-07	1944/2037/6.309/9.42e-07	1443/1524/4.789/9.38e-07
dixmaanf/1500	270/391/0.171/9.23e-07	265/358/0.118/8.35e-07	433/486/0.180/9.08e-07	246/483/0.190/9.56e-07	600/690/0.230/9.78e-07	401/450/0.143/9.78e-07
dixmaanf/3000	290/409/0.256/7.05e-07	359/466/0.294/7.27e-07	534/608/0.383/9.03e-07	287/531/0.334/8.60e-07	787/890/0.561/9.34e-07	698/750/0.474/9.74e-07
dixmaanf/30000	838/1147/6.360/8.98e-07	783/996/5.529/6.94e-07	1804/1950/10.802/9.93e-07	834/1619/8.866/6.56e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaang/300	128/203/0.033/8.32e-07	158/233/0.023/8.75e-07	196/248/0.027/7.09e-07	115/237/0.025/6.40e-07	181/240/0.024/5.21e-07	139/188/0.020/9.94e-07
dixmaang/1500	250/357/0.143/9.69e-07	332/432/0.156/8.96e-07	370/430/0.142/9.74e-07	266/520/0.168/9.22e-07	576/643/0.206/9.39e-07	327/381/0.123/7.25e-07
dixmaang/15000	1609/2151/6.623/4.16e-07	1648/1986/6.108/8.60e-07	1221/1314/4.002/8.31e-07	NaN/NaN/NaN/NaN	1995/2104/6.534/9.60e-07	1016/1079/3.370/7.24e-07
dixmaanh/15000	798/1089/3.437/6.72e-07	719/889/2.708/9.80e-07	1040/1167/3.532/9.25e-07	520/995/3.033/6.81e-07	1768/1856/5.756/9.04e-07	NaN/NaN/NaN/NaN
dixmaanh/21000	613/885/3.599/7.40e-07	1984/4380/17.496/8.67e-07	1436/1573/6.427/9.79e-07	851/1615/6.462/9.00e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanh/30000	874/1178/6.538/8.42e-07	1095/1358/7.782/8.46e-07	1432/1542/8.554/9.78e-07	921/1738/9.501/9.95e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaani/30	217/319/0.032/8.31e-07	203/266/0.007/9.81e-07	421/508/0.014/7.93e-07	202/398/0.012/9.14e-07	422/524/0.015/5.65e-07	257/305/0.009/8.33e-07
dixmaani/90	595/797/0.035/9.48e-07	632/746/0.034/8.92e-07	839/913/0.040/7.41e-07	477/904/0.036/8.24e-07	1632/1732/0.070/8.79e-07	948/1015/0.038/9.27e-07
dixmaani/210	1516/2041/0.123/7.77e-07	1242/1452/0.086/8.82e-07	NaN/NaN/NaN/NaN	1185/2267/0.136/9.99e-07	NaN/NaN/NaN/NaN	1870/1968/0.117/8.96e-07
dixmaanj/6000	858/1161/1.529/7.17e-07	745/925/1.151/6.73e-07	1481/1670/2.108/7.11e-07	592/1081/1.368/8.21e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaank/1500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	1653/3166/1.004/9.56e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanl/300	1113/1523/0.173/8.07e-07	1305/1572/0.156/7.68e-07	NaN/NaN/NaN/NaN	868/1647/0.129/9.20e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanl/21000	910/1234/4.974/6.05e-07	952/1181/4.805/9.08e-07	1282/1467/5.918/8.08e-07	1331/2530/10.192/9.23e-07	1241/1339/5.493/7.61e-07	NaN/NaN/NaN/NaN
dixmaanl/30000	831/1157/6.396/9.53e-07	1141/1394/7.712/9.84e-07	1410/1570/8.696/8.03e-07	674/1299/7.097/8.20e-07	1913/2065/11.569/9.68e-07	NaN/NaN/NaN/NaN
dixon3dq/50	524/735/0.021/5.65e-07	488/623/0.006/9.40e-07	776/862/0.009/7.99e-07	616/1166/0.012/8.84e-07	NaN/NaN/NaN/NaN	493/537/0.006/8.84e-07
dixon3dq/100	936/1260/0.013/9.01e-07	936/1144/0.012/8.27e-07	NaN/NaN/NaN/NaN	821/1576/0.015/9.16e-07	NaN/NaN/NaN/NaN	1158/1235/0.012/9.27e-07
dixon3dq/150	1785/2406/0.023/9.68e-07	1815/2184/0.020/8.29e-07	NaN/NaN/NaN/NaN	1223/2322/0.021/5.56e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dqdrtic/10000	95/226/0.027/6.95e-07	77/228/0.016/5.19e-07	152/258/0.025/3.02e-07	74/222/0.023/9.40e-07	312/457/0.054/7.72e-07	142/209/0.028/2.65e-07
dqdrtic/50000	80/206/0.066/7.68e-07	100/294/0.093/7.26e-07	150/265/0.088/8.12e-07	62/210/0.067/5.14e-07	336/531/0.199/8.33e-07	193/263/0.116/3.59e-07
dqdrtic/150000	95/232/0.285/7.57e-07	76/221/0.264/3.47e-07	149/263/0.343/3.43e-07	86/241/0.282/6.97e-07	296/436/0.709/9.59e-07	138/213/0.357/7.72e-07
dqrtic/5000	54/164/0.175/3.14e-07	56/168/0.172/6.27e-07	62/157/0.158/4.63e-07	41/156/0.165/4.19e-07	187/1226/1.241/9.99e-07	96/185/0.201/3.43e-07
edensch/500	30/105/0.019/2.61e-07	44/195/0.020/8.63e-07	74/399/0.042/9.74e-07	68/474/0.059/3.98e-07	31/84/0.010/9.85e-07	144/941/0.093/9.60e-07
edensch/20000	67/385/1.506/7.93e-07	78/502/1.954/4.01e-07	120/849/3.327/9.79e-07	51/269/1.046/5.85e-07	45/213/0.843/5.72e-07	394/2923/11.501/9.95e-07
edensch/120000	210/1663/36.111/8.37e-07	69/375/8.182/6.99e-07	111/827/17.953/8.11e-07	125/965/20.984/2.12e-07	58/308/6.698/9.15e-07	882/7254/157.550/9.96e-07
eg2/100	236/752/0.017/8.64e-07	303/1422/0.020/7.68e-07	1057/6524/0.087/9.94e-07	538/1585/0.023/4.46e-07	NaN/NaN/NaN/NaN	1086/1250/0.018/8.55e-07
eg2/200	NaN/NaN/NaN/NaN	343/1322/0.028/8.24e-07	NaN/NaN/NaN/NaN	1122/5148/0.104/8.30e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
eg2/500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
fletchcr/7000	139/1147/0.081/4.75e-07	260/2188/0.125/7.02e-07	179/1292/0.070/9.69e-07	391/3583/0.207/9.27e-07	475/4542/0.258/9.60e-07	NaN/NaN/NaN/NaN
freuroth/50	218/1174/0.023/9.11e-07	181/550/0.007/8.05e-07	383/2205/0.028/8.64e-07	236/1585/0.019/9.83e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
freuroth/500	NaN/NaN/NaN/NaN	1087/10723/0.146/9.94e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
freuroth/5000	NaN/NaN/NaN/NaN	923/9512/0.658/7.43e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
genrose/10000	234/392/0.052/5.52e-07	290/462/0.049/9.99e-07	302/418/0.045/8.30e-07	225/473/0.046/7.75e-07	752/877/0.106/9.80e-07	397/475/0.076/5.53e-07
himmelbg/1000	2/13/0.006/7.55e-30	2/13/0.001/7.55e-30	6/24/0.001/2.61e-08	2/13/0.001/7.57e-30	9/17/0.001/1.49e-18	8/12/0.001/1.59e-42
himmelbg/70000	2/15/0.035/8.37e-43	2/15/0.030/8.37e-43	6/26/0.037/5.05e-200	2/15/0.029/8.41e-43	11/42/0.063/0.00e+00	8/15/0.032/9.38e-10
himmelbg/600000	3/16/0.256/5.32e-26	3/16/0.260/1.30e-26	6/37/0.479/1.76e-30	3/16/0.252/8.54e-27	5/17/0.279/4.37e-12	11/19/0.334/8.27e-20
liarwhd/10000	233/680/0.065/3.52e-07	128/456/0.042/3.94e-07	411/823/0.055/3.49e-07	176/628/0.036/5.48e-07	NaN/NaN/NaN/NaN	853/1117/0.136/6.11e-07
liarwhd/30000	143/537/0.115/7.87e-07	164/562/0.122/5.06e-07	459/951/0.210/5.53e-08	201/732/0.150/2.54e-07	NaN/NaN/NaN/NaN	716/994/0.300/2.91e-07
liarwhd/120000	305/882/0.546/3.70e-07	164/651/0.393/1.51e-07	391/836/0.531/5.75e-07	315/1075/0.638/4.67e-07	NaN/NaN/NaN/NaN	679/1018/0.824/8.45e-07
penalty1/100	18/108/0.008/6.16e-07	19/109/0.003/8.13e-07	14/66/0.002/6.61e-07	18/107/0.003/2.82e-07	14/66/0.002/6.62e-07	21/68/0.002/4.28e-07
penalty1/10000	20/94/4.219/2.94e-09	20/92/4.059/4.52e-07	65/281/12.147/4.56e-07	20/91/3.967/8.17e-07	50/259/11.066/6.09e-07	113/180/7.741/3.87e-07
quartc/5000	54/164/0.171/3.14e-07	56/168/0.170/6.27e-07	62/157/0.159/4.63e-07	41/156/0.163/4.19e-07	187/1226/1.248/9.99e-07	96/185/0.201/3.43e-07
tridia/100	361/489/0.010/9.50e-07	251/341/0.004/8.82e-07	809/940/0.009/9.75e-07	353/664/0.007/6.72e-07	1305/1502/0.014/9.35e-07	365/426/0.004/4.46e-07
tridia/300	591/793/0.009/8.11e-07	623/768/0.008/9.58e-07	1702/1959/0.020/9.71e-07	789/1519/0.016/6.12e-07	NaN/NaN/NaN/NaN	797/871/0.009/9.36e-07
tridia/1500	1675/2260/0.038/9.30e-07	1960/2355/0.039/9.73e-07	NaN/NaN/NaN/NaN	1544/2944/0.046/6.80e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
woods/5000	196/534/0.042/7.03e-07	158/434/0.016/8.84e-07	347/476/0.021/6.25e-07	263/640/0.027/7.03e-07	698/1918/0.086/9.63e-07	262/336/0.027/3.07e-07
woods/300000	230/578/1.512/2.66e-07	210/530/1.381/8.38e-07	461/613/1.784/6.53e-07	376/910/2.238/8.20e-07	938/2541/7.108/9.81e-07	324/401/1.507/8.73e-07
woods/1000000	179/436/3.659/4.15e-07	218/648/5.308/5.72e-07	341/493/4.471/9.18e-07	281/702/11.963/7.37e-07	991/2608/47.713/8.60e-07	396/479/11.026/6.24e-07
bdexp/1000	2/11/0.018/2.98e-133	2/11/0.003/2.99e-133	6/25/0.005/1.76e-16	2/11/0.003/3.25e-133	NaN/NaN/NaN/NaN	13/14/0.004/1.41e-11
bdexp/10000	2/17/0.022/2.62e-61	2/17/0.016/2.62e-61	7/21/0.018/7.59e-161	2/17/0.016/2.63e-61	NaN/NaN/NaN/NaN	37/39/0.057/0.00e+00
bdexp/100000	2/12/0.148/4.57e-106	2/12/0.204/4.57e-106	6/22/0.273/5.22e-44	2/12/0.210/4.57e-106	6/22/0.242/8.46e-38	30/37/0.603/0.00e+00
exdenschnf/10000	38/114/0.071/5.03e-07	40/113/0.019/8.83e-07	25/94/0.015/7.73e-07	25/99/0.017/4.94e-08	27/95/0.017/8.85e-07	55/118/0.040/4.22e-07
exdenschnb/3000	25/89/0.016/5.69e-07	26/89/0.008/9.86e-07	28/82/0.007/6.56e-07	19/84/0.006/8.43e-09	25/76/0.005/2.42e-07	50/98/0.008/8.07e-07
exdenschnb/5000000	40/115/7.912/7.22e-07	49/125/9.015/3.21e-07	34/106/7.245/3.53e-07	32/107/7.104/4.32e-07	22/91/6.394/2.64e-07	104/164/15.297/4.03e-07
genquartic/50000	18/87/0.152/5.78e-07	25/93/0.066/3.19e-07	27/89/0.060/1.78e-08	24/116/0.076/3.82e-07	23/85/0.062/5.06e-07	63/117/0.115/9.80e-07
genquartic/100000	19/81/0.095/5.37e-07	21/86/0.091/7.90e-07	32/95/0.102/9.65e-07	19/96/0.097/3.50e-07	30/96/0.110/5.54e-07	75/128/0.193/3.92e-07
biggsb1/250	1282/1688/0.071/7.82e-07	1518/1838/0.081/9.91e-07	1912/2019/0.085/7.25e-07	1362/2593/0.115/9.63e-07	NaN/NaN/NaN/NaN	1737/1820/0.109/6.18e-07
biggsb1/500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
fletcbv3/100	873/1724/0.151/9.53e-07	986/2159/0.166/7.87e-07	1037/1324/0.112/5.49e-07	1472/2974/0.271/2.43e-07	NaN/NaN/NaN/NaN	648/833/0.083/8.72e-07
nonscomp/10000	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	56/122/0.019/8.83e-07	NaN/NaN/NaN/NaN	50/116/0.020/9.56e-07	144/207/0.068/3.38e-07
nonscomp/70000	NaN/NaN/NaN/NaN	1774/2707/2.849/8.13e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	95/158/0.202/9.30e-07
nonscomp/100000	NaN/NaN/NaN/NaN	1831/2767/3.552/8.60e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	298/368/0.636/3.68e-07
power1/100	1137/1545/0.060/7.45e-07	1360/1617/0.063/9.75e-07	NaN/NaN/NaN/NaN	1042/1975/0.106/8.32e-07	NaN/NaN/NaN/NaN	1757/1870/0.095/4.54e-07
raydan1/300	137/205/0.025/5.48e-07	173/249/0.018/9.08e-07	299/370/0.023/4.73e-07	164/312/0.016/4.36e-07	171/223/0.012/6.76e-07	178/228/0.012/9.49e-07
raydan2/2000	14/71/0.016/7.00e-07	14/71/0.009/7.00e-07	14/65/0.009/9.35e-07	14/71/0.010/7.00e-07	14/64/0.009/2.58e-07	33/73/0.014/3.59e-08
raydan2/70000	21/102/0.496/2.05e-07	12/72/0.457/2.93e-07	21/75/0.501/1.28e-07	30/159/1.068/1.59e-07	47/103/0.831/3.02e-07	86/132/1.193/7.24e-07
diagonal1/250	1066/8565/0.624/8.36e-07	550/4644/0.366/8.94e-07	789/6148/0.512/9.14e-07	449/3517/0.285/9.72e-07	627/5285/0.483/9.91e-07	NaN/NaN/NaN/NaN
diagonal2/500	156/268/0.034/4.95e-07	158/264/0.034/9.84e-07	310/372/0.031/9.53e-07	154/300/0.022/5.88e-07	666/1789/0.149/8.07e-07	223/265/0.037/8.42e-07
diagonal2/5000	463/741/0.321/9.56e-07	535/842/0.377/7.01e-07	1106/1235/0.582/8.59e-07	470/922/0.413/8.23e-07	NaN/NaN/NaN/NaN	891/945/0.584/9.90e-07
diagonal2/10000	764/1191/0.972/9.71e-07	696/1043/0.858/5.81e-07	1750/2011/1.658/9.94e-07	689/1354/1.061/7.95e-07	NaN/NaN/NaN/NaN	1152/1229/1.250/9.81e-07
singx/300	307/812/0.205/5.69e-07	161/438/0.116/9.85e-07	725/1059/0.205/8.92e-07	286/671/0.112/5.32e-07	NaN/NaN/NaN/NaN	966/1160/0.197/5.77e-07
singx/1000	225/528/1.837/4.63e-07	243/620/2.084/7.79e-07	895/1307/5.209/7.18e-07	633/1489/4.967/4.88e-07	NaN/NaN/NaN/NaN	610/746/2.676/4.90e-07
singx/3000	188/460/12.540/7.87e-07	347/769/19.403/4.80e-07	618/912/23.152/9.37e-07	688/1701/42.846/4.29e-07	NaN/NaN/NaN/NaN	1243/1510/18.791/8.91e-07
lin/30	17/79/0.023/6.58e-07	17/79/0.012/6.96e-07	19/67/0.006/2.67e-07	17/79/0.006/6.96e-07	19/67/0.005/2.57e-07	17/61/0.004/3.28e-07
lin/400	14/72/0.144/7.97e-07	14/72/0.151/7.97e-07	16/66/0.118/8.77e-07	14/72/0.127/7.97e-07	16/66/0.116/8.77e-07	38/83/0.144/3.76e-07
lin/700	18/81/4.536/2.20e-07	18/81/4.560/2.85e-07	29/90/4.994/1.85e-07	18/81/4.488/2.20e-07	29/90/4.981/6.48e-08	40/86/4.773/4.06e-07
osb2/11	606/1011/0.044/7.30e-07	414/708/0.023/8.22e-07	1626/1924/0.064/9.32e-07	581/1123/0.030/7.27e-07	NaN/NaN/NaN/NaN	545/621/0.013/9.13e-07
pen1/50	136/510/0.015/7.03e-07	67/315/0.005/8.38e-07	145/493/0.007/9.90e-07	94/417/0.007/1.44e-07	663/3305/0.047/8.81e-07	279/429/0.006/5.92e-07
pen1/100	172/555/0.011/7.82e-07	60/329/0.006/8.95e-07	393/1605/0.028/2.98e-07	169/636/0.011/2.24e-07	1665/8417/0.135/9.94e-07	383/621/0.010/2.13e-08
pen1/600	161/1025/1.723/9.99e-07	117/478/0.802/9.98e-07	207/756/1.260/5.40e-07	87/548/0.912/7.68e-07	NaN/NaN/NaN/NaN	304/437/0.737/8.26e-07
pen2/10	1245/3208/0.056/7.65e-07	846/2413/0.035/7.92e-07	NaN/NaN/NaN/NaN	1807/5105/0.059/8.44e-07	NaN/NaN/NaN/NaN	1460/1891/0.022/7.81e-07
pen2/50	381/1191/0.023/9.45e-07	287/921/0.018/6.02e-07	1156/2033/0.039/7.75e-07	424/1318/0.026/7.50e-07	1193/6006/0.117/9.05e-07	1397/1638/0.032/9.97e-07
pen2/90	227/755/0.030/8.46e-07	207/684/0.026/7.77e-07	603/1109/0.043/9.05e-07	212/587/0.023/9.21e-07	993/4155/0.157/9.44e-07	882/3544/0.129/9.89e-07
rosex/300	104/296/0.039/9.51e-07	107/292/0.022/9.86e-07	184/365/0.031/5.60e-07	95/323/0.026/9.07e-07	586/2399/0.161/9.59e-07	136/221/0.015/4.15e-07
rosex/1000	99/348/1.017/9.04e-08	107/340/0.984/9.72e-07	210/381/1.103/7.20e-07	114/359/1.053/5.02e-07	669/2877/8.378/9.86e-07	448/542/1.527/9.97e-07
trid/400	86/148/0.084/7.24e-07	85/145/0.061/6.10e-07	41/93/0.047/9.71e-07	99/191/0.087/8.01e-07	37/91/0.053/6.92e-07	48/103/0.048/4.59e-07